

Homework (Carolina Reaper)

- Q1.** A painter mixes 3 parts blue paint to 5 parts white paint. What is the ratio of blue to total paint?
(A) 3 : 8
(B) 5 : 8
(C) 3 : 5
(D) 8 : 3
(E) 5 : 3
- Q2.** A DJ mixes music with a rhythm pattern of 2 slow beats to 5 fast beats. What is the proportion of slow to total beats?
(A) 2 : 7
(B) 5 : 7
(C) 2 : 5
(D) 7 : 2
(E) 5 : 2
- Q3.** An artist creates a canvas with a width to length ratio of 3 : 5. If the width is 9 inches, what is the length?
(A) 15 inches
(B) 12 inches
(C) 20 inches
(D) 25 inches
(E) 30 inches
- Q4.** A music frequency has a ratio of 2 : 3 for two consecutive notes. If the first note has a frequency of 200 Hz, what is the frequency of the second note?
(A) 100 Hz
(B) 300 Hz
(C) 250 Hz
(D) 400 Hz
(E) 500 Hz
- Q5.** A sculptor creates a model with a 1 : 4 ratio of height to base diameter. If the base diameter is 16 inches, what is the height?
(A) 4 inches
(B) 8 inches
(C) 12 inches
(D) 16 inches
(E) 20 inches
- Q6.** A sound engineer mixes two audio tracks with a 3 : 2 ratio of treble to bass. If the treble level is 12 decibels, what is the bass level?
(A) 8 decibels
(B) 10 decibels
(C) 12 decibels
(D) 16 decibels
(E) 20 decibels
- Q7.** A painter uses a 5 : 3 ratio of primary to secondary colors. If 15 gallons of primary colors are used, how many gallons of secondary colors are used?
(A) 9 gallons
(B) 12 gallons
(C) 15 gallons
(D) 18 gallons
(E) 20 gallons
- Q8.** A musician composes a piece with a 4 : 3 ratio of quarter notes to eighth notes. If there are 32 quarter notes, how many eighth notes are there?
(A) 12 eighth notes
(B) 16 eighth notes
(C) 20 eighth notes
(D) 24 eighth notes
(E) 28 eighth notes

Open Ended Questions (FRQ)

- Q9.** A photographer takes a picture with a 3 : 2 aspect ratio. If the width of the picture is 24 inches, what is the length? Show your calculation and express your answer as a ratio and a decimal value.
- Q10.** A dancer performs a routine with a 5 : 2 ratio of fast to slow steps. If the dancer takes 50 fast steps, how many slow steps does the dancer take? Show your calculation and express your answer as a ratio and a whole number value.

Chef's Tips

- Tip 1: Encourage students to use cross-multiplication for proportion problems.
- Tip 2: Remind students to simplify their ratios and proportions.
- Tip 3: Emphasize the importance of showing calculations and expressing answers in multiple forms.